

AI-Powered Municipal Solid Waste & Circular Economy Platform

Problem Statement

Introduction

Rapid urbanization and increasing population growth have created significant challenges in municipal solid waste management. Many cities and municipalities face difficulties in efficient waste collection, waste segregation, resource allocation, and monitoring of waste management activities.

Traditional waste management systems often depend on manual processes and lack intelligent data-driven decision-making capabilities. This results in inefficient collection routes, increased operational costs, improper waste segregation, and limited understanding of recycling and circular economy opportunities.

Existing Challenges

Municipal waste management faces several challenges:

- Inefficient planning of waste collection routes.
 - Lack of real-time analysis of waste generation patterns.
 - Poor waste segregation awareness among citizens.
 - Difficulty in tracking recycling and sustainability performance.
 - Limited use of Artificial Intelligence for waste management decisions.
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Proposed Solution

The **AI-Powered Municipal Solid Waste & Circular Economy Platform** aims to provide an intelligent decision-support system for municipalities and waste management organizations.

The platform uses Artificial Intelligence and IBM Granite technology through watsonx.ai to analyze waste-related information and generate useful recommendations for improving waste management operations.

The system focuses on improving collection efficiency, supporting circular economy practices, and enabling data-driven decision-making.

Objectives

The main objectives of this project are:

- To analyze municipal solid waste generation patterns.
 - To optimize waste collection planning and resource utilization.
 - To provide AI-based recommendations for better waste management.
 - To support circular economy analytics and sustainability goals.
 - To improve decision-making using Artificial Intelligence.
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AI-Based Approach

The project uses IBM Granite AI capabilities to provide intelligent insights and recommendations.

The AI-powered platform helps in:

- Waste management analysis.
 - Collection optimization.
 - Sustainability monitoring.
 - Generating data-driven suggestions.
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Domain

Urban Waste Management

Technologies Used

- Artificial Intelligence
 - IBM Granite AI Model
 - IBM watsonx.ai
 - Python
 - Flask
 - HTML5
 - CSS3
 - JavaScript
 - Bootstrap
 - SQLite
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Expected Outcome

The proposed platform provides a smart and efficient approach for municipal solid waste management by combining Artificial Intelligence with circular economy principles.

The system helps municipalities improve operational efficiency, reduce resource wastage, and support sustainable waste management practices.